

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0182

East Danville 100 kV
September 15, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0182. The Interconnection Customer proposes a 240.0 MW natural gas combined cycle facility interconnecting at the East Danville 100 kV substation in the Illinois area. The request was received on December 05, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 120 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,002,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PICKFORD to HERALD-TWIN CITIES 230.0 kV	TREGO SUBSTATION to BEAVER 500.0 kV	102.4	664	10.5
KITSAP to E DANVILLE 100.0 kV	E DANVILLE to CHURCH ROCK TAP 100.0 kV	102.2	113	59.8
HIGHGATE to HERALD-TWIN CITIES 230.0 kV	MASON ST 500/230kV Auto	101.8	518	7.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HERALD-TWIN CITIES to PICKFORD 230.0 kV Line	Reconductor line	\$13,963,000	5% (\$636,000)
KITSAP to E DANVILLE 100.0 kV line	Reconductor line	\$1,922,000	100% (\$1,922,000)
HIGHGATE to HERALD-TWIN CITIES 230.0 kV	Reconductor line	\$9,754,000	5% (\$444,000)
	Total allocated to project		\$3,002,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.